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Assessing Lockdown Impacts on PM. Levels: Evidence from Italy and Japan

Causal Inference from Matched City-Level Data

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Abstract — The paper sets out a rigorous econometric assessment that attempts to evaluate the causal effect of COVID-19 lockdown measures on urban air quality, specifically looking at concentrations of PM2.5. Using a Difference-in-Differences (DiD) framework with city-level Propensity Score Matching (PSM), the study compares Italy, known for its strident and severe restrictions, with Japan, which followed comparatively moderate restrictions. This is presented in three approaches: first elaborately the aggregated country-level analysis, further detailing the matched baseline city-day DiD model, then performing a robustness consideration with one-to-one-matched cities. Substantial evidence across various considerably credible robustness tests—placebo tests, parallel trend validations, corrections for heteroscedasticity via robust standard errors, dynamic city-level matching—reiterated and solidified the major conclusions from the analysis in regards to a reduction of PM2.5 concentrations due to lockdown measures. This also makes the estimated effects very interesting since along the division lines of our heterogeneous findings, one should note that more urbanized Italian cities made further air quality improvements, a finding we interpret as supporting a key role for localized economic and demographic factors when mediating the impacts of lockdown. Findings allow for an analysis of the massive comparative environmental gains made through the cutting of key industrial and transportation activities with clear recommendations to be circulated to policymakers that find the point to weigh up economic growth, public health needs, and environmental sustainability measures. The study contributes to the existing literature through dynamic weather adjustments and fine matching analyses and maximizes the precision and political significance of such estimated causal impacts. The findings argue optimally combined policy measures to promote economic and environmental goals, drawing critical implications for urban planning, environmental sustainability strategies, and public health policy.

1 Introduction

Due to the COVID-19 pandemic, there were giant stress examples for economic activity and human mobility around the globe, providing a unique natural experiment to study the causal impact of human activities on air quality. With lockdowns being enforced all over the world, a sharp drop in emissions of different air pollutants did yield quite some insights on how changes in mobility and industrial activity have a role in the level of pollution. PM2.5 in the atmosphere, or airborne particulate matter with a diameter smaller than 2.5 micrometers, has particular significance among pollutants due to its adverse health consequences, respiratory and cardiovascular diseases being just some of them. Notable PM2.5 decline during lockdown has been noted because its fine particles are almost entirely generated as a result of human activity, including industrial ones, transport, and energy generation.In this study, by focusing on both Italy and Japan, we will pay special attention to contrasting lockdown policies during the COVID-19 pandemic. While Italy imposed strict national lockdown measures starting late February 2020, thereby heavily limiting mobility and industrial activity, Japan, on the other hand, pry for a range of relatively weak and less restrictive approaches dependent greatly upon voluntary compliance with limited closures. Hence, this contrast is a perfect ground for a meaningful exploration of causal impacts of lockdown stringency on air pollution level.To causally assess the effect of lockdown stringency on air quality, we combine the Difference-in-Differences (DiD) methodology with Propensity Score Matching (PSM). This procedure relies on a matching strategy that pairs each Italian city (treatment group) with a sufficiently similar city in Japan (control group) according to the key economic and demographic variables, namely GDP per capita and log population. GDP per capita is an important economic characteristic since it is indicative of the levels of economic activities and prosperity that those cities manage to achieve, which directly de-

termines the level of industrial emissions and, hence, the level of pollution. Population size signifies potential disparities in human activity, urban dynamics, and initial levels of pollution intensity.

After pairing the two groups according to their characteristics, we examined changes in daily PM2.5 concentrations before and after implementation of lockdown. The analysis uses fine-grained meteorological covariates—temperature, humidity, atmospheric pressure—to control for weather-related variability in pollution concentrations and thus isolate the causal effect of the lockdown measures. The initial country-level aggregated analysis gives the preliminary sense while the main model thus depends on the matched set containing detailed city-day level information to yield robust and accurate causal inference. To ensure the identified effects were consistent and unaffected by random short-term fluctuation or variability in matching, robustness checks included one-to-one city-level matching. Two countries are studied: Italy and Japan, which adopted almost opposite lockdown policies during the pandemic. Stringent lockdown of the economy and society began in Italy by late February 2020, having massive implications on economic and social activities. In contrast, Japan took a softer approach that relied more on voluntary compliance and less on legally enforceable restrictions. These differences provide a unique chance to disentangle the effects of tough lockdown measures on PM2.5 levels, especially in urbanized regions where baseline pollution levels and industrial activities are higher.

2 Literature Review

The COVID-19 pandemic offered a unique opportunity to investigate how human activities impacted air quality, consequently shedding light on the environmental implications of lockdown restrictions around the world. One of the main contributions in the field was by Ma et al. (2024), who addressed the air quality impacts brought about by lockdowns in China, the USA, Italy, and Germany, applying meteorological normalization and DiD methods. Results revealed reductions of NO2 and PM2.5, regional variations in PM2.5, an indication of health benefits subsequent to air quality improvements, and the need for customized regional policy interventions. Satellite-based assessments further enhanced this understanding. Levelt et al. (2022) demonstrated pronounced declines in global NO2 (14%-63%) and SO2 levels during lockdowns, attributed to reductions in transportation and industrial activities.

These findings revealed substantial evidence of spatial and temporal variability in air quality changes, stressing the value of satellite observations in complementing ground analyses and emission trends monitoring. Nguyen et al. (2022) in Vietnam also studied the relationship between air quality and lockdown stringency, reporting major declines in CO values. Stricter activity restrictions were shown to have larger paybacks in terms of the environment, with more considerable reductions in more actively economically engaged areas. Coinciding with the present study's aim to understand the causal impacts of lockdowns on PM2.5 in Italy and Japan, a systematic review by Dahu et al. (2022) synthesized evidence from other studies all over the globe indicating collective reductions in NO2, CO, PM2.5, and PM10 throughout lockdowns. The same review found an association between lower levels of pollution and less severe COVID-19 and mortality, lending support to the implications pollution control has on public health. Addas and Maghrabi (2022) undertook a global review of 144 studies to reveal notable improvements in air quality during the lockdown periods. This work brought toward light substantial reductions in PM2.5, PM10, NO2, and CO levels with Asia and Europe being the most widely studied areas. These findings shed light upon the degrading environment when urban centers witness reduction in human-activity. Liu et al. (2021) note huge declines in global pollutant levels during the pandemic: NO2 (23-37%), PM10 (14-20%), SO2 (2-20%), PM2.5 (7-16%), and CO (7-11%). However, a rise in ozone (10-27%) was noted due to atmospheric chemistry. Their analysis estimated that due to improved air quality during the pandemic, up to 146,649 premature deaths were averted worldwide, thus demonstrating the potential of restrictions on mobility to yield considerable health and ecological gains. This set of studies presents concrete evidence of the varying environmental approaches to the COVID-19 lockdown, along with a demonstration of major reduction in air pollution and inherent health benefits. These findings would also serve as ground for this study as it brings in matching methodological techniques, which have been utilized to estimate lockdowns' causal effects on PM2.5 at the city level.

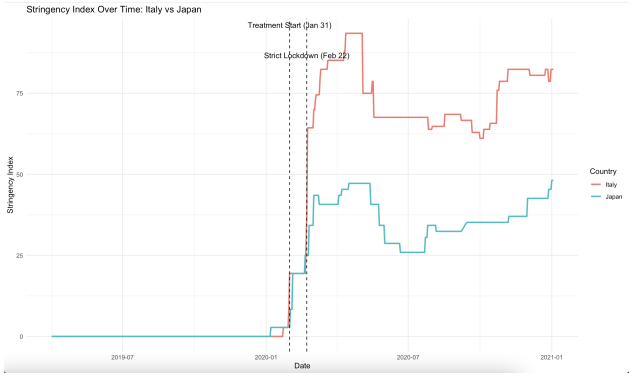


Figure 1 Stringency Index Over-Time

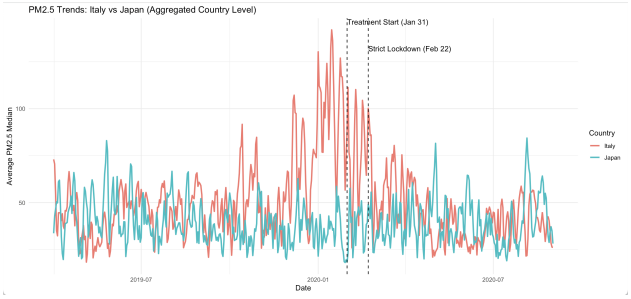


Figure 2 Aggregated Country Level Data

3 Data Sources

This study evaluates the causal impact of COVID-19 lockdown measures on PM2.5 concentrations, using daily city-level data from Italy and Japan spanning April 1, 2019, through September 1, 2020. Two primary datasets are employed.

3.1 Air Quality Open Data Platform

The air quality and meteorological datasets used in this analysis were obtained from the World Air Quality Index (WAQI) project via the Air Quality Open Data Platform. WAQI currently provides an extensive and easily accessible, 3 times-a-day updated global dataset since January 2015 on input regarding near to about 380 major cities from all over the globe. For any city, the datasets provided represent daily median values coming in from the several air quality measuring stations. The dataset contains pollutant species PM2.5, PM10, Ozone and many more, according to US EPA standards as well as meteorological inputs like temperature, humidity, wind speed, wind gust, atmospheric pressure, and dew point. All reported times are in the Coordinated Universal Time (UTC). The daily city-level median values were made using all available station observations to reduce biases from missing and extreme values.

3.2 Oxford COVID-19 Government Response Tracker (OxCGRT)

Lockdown policy severity is represented as the Stringency Index taken from the Oxford Coronavirus Government Response Tracker (OxCGRT) through Our World in Data. The index pooled nine response metrics: school closures, workplace closures, cancelation of public events, restrictions on public gatherings, closure of transport systems, stay-at-home orders, public information campaigns, restrictions on movement within countries, and international travel controls. They were each scored on a scale of 0-100, with a daily mean index calculated. The higher the score, the stricter the policy measure (100 is the most stringent). Although there are some sub-national variations, the OxCGRT index informs a country about the toughest measure taken for all the cities under it, providing some consistency in cities falling within the same country's territories.

4 Methodology

4.1 Matching

Table 1 Matched City Pairs for City-Day Level Matching

Italy City	Japan City	Italy Log Pop.	Japan Log Pop.	Italy Log GDP	Japan Log GDP
Bologna	Utsunomiya	13.83	13.15	10.78	10.74
Brescia	Nagano	14.04	12.81	10.62	10.59
Florence	Osaka	13.81	14.83	10.70	10.68
Livorno	Chiba	12.70	13.80	10.38	10.38
Milan	Tokyo	14.99	16.47	11.13	11.21
Modena	Toyama	13.46	12.94	10.76	10.69
Naples	Nara-shi	14.91	12.81	10.05	10.19
Parma	Utsunomiya	13.02	13.15	10.78	10.74
Prato	Sapporo	12.47	14.49	10.48	10.48
Rome	Kanazawa	15.26	13.05	10.69	10.60
Trieste	Toyama	12.34	12.94	10.69	10.69
Turin	Takamatsu	14.61	12.93	10.57	10.56
Bologna	Shizuoka	13.83	13.43	10.78	10.73
Trieste	Osaka	12.34	14.83	10.69	10.68
Florence	Kanazawa	13.81	13.05	10.70	10.60
Brescia	Hiroshima	14.04	13.99	10.62	10.59
Rome	Nagano	15.26	12.81	10.69	10.59
Bologna	Nagoya	13.83	14.66	10.78	10.86

Note: This table presents the matched city pairs used in the city-day level Difference-in-Differences analysis. The matching was performed based on log-transformed population size and GDP per capita.

The air quality data were preprocessed to clean the missing data before the analysis. Specifically, meteorological observations that were missing were substituted by city mean daily values. To increase comparability between the treatment (Italian) and control (Japanese) cities, two additional city-level socioeconomic variables, log GDP per capita and log popula-

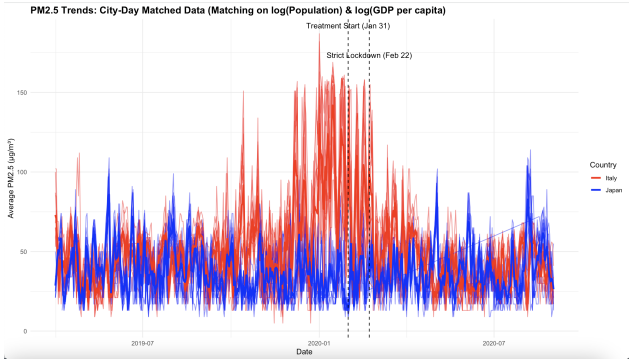


Figure 3 PM2.5 trends for city-day matched data

tion, were added. The cities' GDP per capita figures (in euros, originally from 2021) for the Italian ones were converted to U.S. dollars in 2018 by first deflationary standardization via GDP deflators (109.6 and 114 for 2018 and 2021, respectively) and secondly converted to 2018 dollars using the prevailing official exchange rates. This adjustment ensured that the GDP per capita data for the Japanese cities in 2018 (reported in U.S. dollars, thus, PPP-adjusted) is consistent and comparable economically.

In terms of matching Italian and Japanese cities, two approaches were applied in regard to matching specifications of log population and log GDP per capita. City cities in Japan and Italy were matched daily using the Propensity Score Matching (PSM) method, with a 1:1 nearest neighbor matching. The daily matching principle allows assessing robustness while adjusting for adverse short-term/temporal changes of matched city pairs. Cities were therefore also matched at an aggregate level, stabilizing city pairs during the entire analysis period as another robustness check. Well-matched pairs included urban centers such as Milan-Nagano and Rome-Tokyo, enabling direct comparison and further verifying stability of results.

The covariate balance diagnostics which proved effective through computing standardized mean differences, empirical cumulative density functions, and visual inspections by using Love plots, provided sufficient evidence that matching was quite successful in reducing potential covariate imbalance.

4.2 Analytical Framework (Difference-in-Differences)

Manifestation of the lockdown effects is understood through the DiD methodology: Italy is the treatment group (high-stringency lockdown), while Japan is the control (low-stringency lockdown).The date is set as January 31, 2020.'s treatment start date' coincides with the point in time when incipient policy responses set in; ultimately, this enables an assessment of policy impacts from the window of opportunity these nascent policy interventions presented, yet the most stringent lockdown measures in Italy did not happen until February 22.2020.

Undaunted, the impossibility of approaching the outcome through country-level data was bypassed through DiD in order to measure trends of PM2.5 cc over time and distance, setting notably the first pronouncements about possible differences in concentrations between Italy and Japan.An initial aggregate country-level analysis with DiD further helped explore someanimum time-and-space trends of PM2.5 concentrations in Italy and Japan.Thus, while providing some concerning overall trends regarding air quality, this analysis was only proper initiative towards a planned matched detail city-level analytical discourse performed subsequently, since it aggregates country-level city data-level data medians into daily median computations subsequent to such periodical exploration of city-specific pairing frameworks based on developed-cut- spatial-matching statistics which upheld towards publication based on positive predictive power supports—city-day matched and fixed and constant bound numeration sequential city pair random developing allocations—the base primary analysis the robust verification.Thus, city-day matching for treatment-impact assessment pairs cities each day on log population and log GDP per capita similarity, establishing the first analytical framework to identify lockdown impact; furthermore, 1-1 city management has double-checked the baseline outcomes.

The DiD estimation accounts for the interaction term (Treatment × Post), which explicitly captures lockdown-driven changes in PM 2.5. The empirical model estimates city fixed effects, which control for unobserved, city-specific, time-invariant characteristics, and incorporates monthly fixed effects to account for seasonality in the data in order to control for other sources of variation. The inclusion of meteorological covariates was meant to disentangle the lockdown-induced effect from confounding effects due to weather variability: temperature, humidity, wind speed, wind gust, and atmospheric pressure, though dew point was dropped from the final models due to its high multicollinearity with temperature, as judged through correlation diagnostics and VIF tests.

4.3 Robustness Checks and Heteroscedasticity Robust Inference

Heteroscedasticity checks were performed on DiD results to improve confidence by using the Breusch-Pagan test, revealing evidence of heteroscedasticity in the residuals for the reduced and full fixed-effects models. Hence, HC1 type robust standard errors were applied in order to achieve valid statistical inference and hypothesis tests for the estimates of DiD. Robust inference results indicated persistent significant and negative interaction terms across models, supporting the notion that stricter lockdown measures reduced PM 2.5 levels. The robustness checks permitted enhancement of confidence in the study's causal conclusions and allowed the verification of results against possible violations of classical regression assumptions.

In summary, the methodological approach of this study combined rigorous matching techniques, thorough DiD assessments at both city and aggregate levels, and comprehensive robustness tests. Drawing on well-matched pairs of cities, robust statistical inference, and systematic controls for possible confounders, the current analysis provides evidence that is credible regarding the causal impact of lockdown measures due to COVID-19 on the improvement of urban air quality. The findings provide crucial intelligence for decision-makers assessing the environmental consequences of public health interventions amid a delicate balancing act between economic activity, public health goals, and sustainable development.

5 Model Specification

The general specification for the Difference in Differences (DiD) model used in this study is as follows:

$$\begin{aligned} PM2.5_{it} = \beta_0 &+ \beta_1 \text{treatment}_i + \beta_2 \text{post}_t \\ &+ \beta_3 (\text{treatment}_i \times \text{post}_t) \\ &+ \gamma \text{Controls}_{it} + \epsilon_{it} \end{aligned} \tag{1}$$

Here, $PM2.5_{it}$ represents the daily median $PM2.5$ concentration for city i at time t. The variable treatment_i is a binary indicator (equal to 1 if city i is in Italy and subject to strict lockdown measures, 0 if in Japan), and post_t is a binary indicator defining the post-treatment period (equal to 1 if the date t is on or after January 31, 2020, when initial policy responses began). The interaction term ($\text{treatment}_i \times \text{post}_t$) represents the DiD estimate, measuring the lockdown's causal impact on $PM_{2.5}$ levels.

The vector Controls_{it} includes daily meteorological covariates that influence air quality, specifically temperature, humidity, wind speed, wind gust, and atmospheric pressure. Dew point is excluded due to multicollinearity with temperature. For improved estimation accuracy, fixed-effects models incorporate city and month fixed effects as follows:

$$\begin{aligned} PM2.5_{it} = \beta_0 &+ \beta_1 \text{post}_t + \beta_2 (\text{treatment}_i \times \text{post}_t) \\ &+ \gamma \text{Controls}_{it} + \delta_{City_i} + \lambda_{Month_t} + \epsilon_{it} \end{aligned} \tag{2}$$

where δ_{City_i} represents city fixed effects that control for all unobserved, time-invariant characteristics at the city level (including the treatment variable, which is omitted since it does not vary over time and thus is absorbed), and λ_{Month_t} captures monthly fixed effects to control for seasonal patterns affecting air quality.

Given the serious heteroscedasticity that preliminary diagnostics indicated (Breusch-Pagan tests), HC1 type robust standard errors were used in all DiD regressions to make sure the statistical inference is valid. Such a model controls for confounding factors and effectively narrows the change in air quality to the lockdown itself, yielding estimates of the lockdown causal effect on $PM_{2.5}$ concentrations that are stable and credible.

6 Model Diagnostics

This multivariate diagnosis involved several tests for the assurance of validity and reliability in our econometric analysis: mostly on multicollinearity issues, covariate balance, and the parallel trend assumption.

Table 2 Variance Inflation Factor (VIF) for Aggregated Country-Level Data

Variable	VIF
Treatment	2.77
Post	2.76
Interaction (Treatment × Post)	3.49
Temperature	12.04
Humidity	2.63
Wind Speed	1.68
Wind Gust	2.07
Pressure	2.11
Dew Point	13.75

Note: Variance Inflation Factors (VIF) measure multicollinearity. Higher values (>10) suggest high collinearity among predictors.

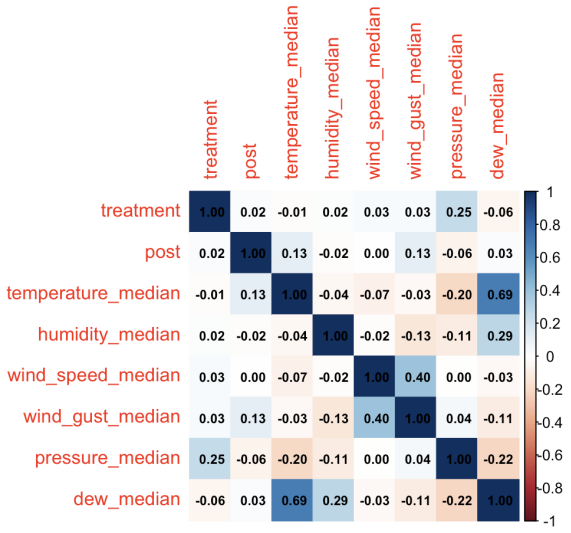


Figure 4 Correlation Matrix for Predictor Variables in Matched City-Day Level Data

6.1 Multicollinearity and Correlation Analysis

A pairwise correlation matrix and its VIF scores showed existence of multicollinearity between VIF values for the aggregated model country-based and matched city-day level DiD models. In the simple country-level analysis, there had been very high correlation coefficients of temperature and dew point at 0.91, hence VIF rose exceptionally high (13.75 for dew point and 12.04 for temperature). It was for concerns of multicollinearity that dew point was dropped from subsequent models of the aggregated level across countries.

Table 3 Variance Inflation Factor (VIF) for Matched City-Day Level Data

Variable	VIF
Treatment	1.67
Post	2.14
Temperature	2.25
Humidity	1.31
Wind Speed	1.20
Wind Gust	1.25
Pressure	1.14
Dew Point	2.40
Interaction (Treatment × Post)	2.75

Note: Variance Inflation Factors (VIF) measure multicollinearity. All values remain below the critical threshold of 10, indicating no severe multicollinearity issues.

6.2 Matching Diagnostics and Covariate Balance

At the matched city-day level, the correlation between temperature and dew point was also high (0.69). The VIF scores for the matched city-day models, however, were lower than those for the aggregated models (temperature VIF = 2.25; dew point VIF = 2.40), and dew point was thus excluded again to ensure consistent treatment across models and straightforward interpretation of coefficients. Following the elimination of dew point, the concerns about multicollinearity went away, with all the remaining covariates now bearing VIF scores within the acceptable limit of 2.8.

Table 4 Matching Diagnostics and Covariate Balance

Variable	Means Treated	Means Control	Std. Mean Diff.	Var. Ratio	eCDF Mean	eCDF Max	Std. Pair Dist.
Before Matching (All Data)							
Distance	0.3043	0.2592	0.4364	1.6820	0.1929	0.4848	–
Log Population	13.7973	13.6169	0.1866	1.1858	0.1034	0.2279	–
Log GDP per Capita	10.6336	10.5420	0.3668	1.9131	0.2017	0.4848	–
After Matching (Matched Data)							
Distance	0.3043	0.2975	0.0664	0.9511	0.0501	0.2446	0.2117
Log Population	13.7973	13.7517	0.0472	0.8321	0.1114	0.2449	1.3237
Log GDP per Capita	10.6336	10.6240	0.0384	1.1183	0.0562	0.2446	0.2110
Sample Sizes							
Control (All)	16,401	Treated (All)	6,112	Matched Control	6,112	Matched Treated	6,112
Unmatched	10,289	–	–	–	–	–	–
Discarded	0	–	–	–	–	–	–

Note: The table presents the balance diagnostics before and after matching. Standardized mean differences, variance ratios, and empirical cumulative distribution function (eCDF) differences are provided for key covariates. A successful match is indicated by reductions in standardized mean differences and eCDF discrepancies post-matching.

Using nearest-neighbor propensity score matching (PSM) on log population and log GDP per capita, the matching procedure established a comparability among Italian (treated) and Japanese (control) cities. Matching quality was assessed by exploring standardized mean differences (SMD) and empirical cumulative density functions (CDF) before and after matching. Matching variables showed large discrepancies before matching. The absolute SMD values of distance and log population were 0.44 and 0.19, respectively, which are meaningful differences. Following matching, such differences were reduced significantly (i.e., SMD of distance dropped to 0.066, log population dropped to 0.05, and log GDP per capita to 0.056), indicating successful balance achieved over the matched city pairs. Successful balancing of covariates allows for cleaner comparisons in the subsequent DiD analysis while minimizing potential biases due to differences in demographic and economic characteristics.

6.3 Parallel Trend Assumption

Inference of parallel trends assumption about the methodology of DiD, the visual and statistical com-

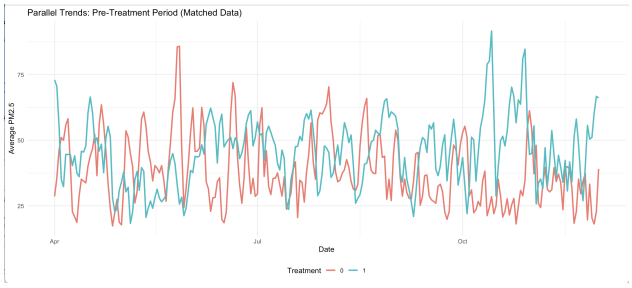


Figure 5 Parallel Trend Plot

Table 5 Difference in $PM_{2.5}$ (Italy - Japan) During Pre-Treatment

Statistic	Mean	SD	Min	Max
Difference	7.25	19.7	-60.2	63.2

Note: This table presents the mean, standard deviation (SD), minimum, and maximum values of the difference in $PM_{2.5}$ levels between Italy and Japan during the pre-treatment period.

parison of trends for PM2.5 was between two matched sets of Italian and Japanese cities gone before treatment (April 1, 2019 -January 30, 2020). It was confirmed the overlapping trends of the two groups so-called pre-treatment showing common pattern with very slight short-term seasonal variation thereby making it impossible for any substantial diverging or unique difference of PM2.5 path between the groups. The average difference in PM2.5 levels between treated and control cities during the pre-treatment period was moderate (mean difference= 7.25 $\mu g/m^3$, standard deviation= 19.7 $\mu g/m^3$), which additionally indicated support of the parallel trends assumption on which the DiD methodology hinges.

6.4 Placebo Test (Falsification Check)

In support of the reliability and plausibility of the Difference-in-Differences estimates, a placebo test (falsification check) was carried out. The aim was to

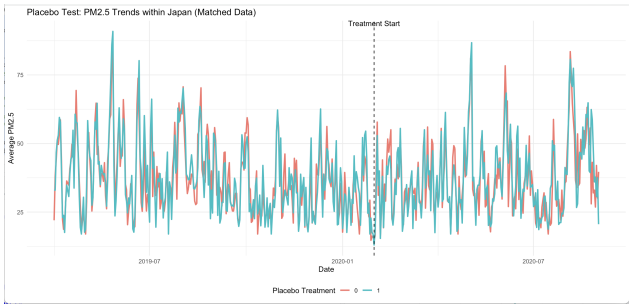


Figure 6 Placebo Test

determine whether it could be that the DiD estimates were created due to random events, thereby confirming that no actual lockdown intervention existed. The design of the placebo analysis allows one to consider only cities in the original control group (Japanese cities), in which there are no real treatment effects imposed by lockdown policies. A nonreal placebo attributable indicator was randomly assigned to about half of the control cities, devising an imaginary treatment completely disconnected from real lockdowns. In this case, the interaction term captures merely random perturbations and not actual effects from the policy. The placebo regression model was specified similarly to the parent DiD model, additive meteorological control variables (temperature, humidity, wind speed, wind gust, and atmospheric pressure). The results indicated that the placebo interaction term was insignificant statistically with (-0.249) estimate and (0.760) p-value, confirming the absence of any artificial effect. Such a finding is important for effectively ensuring that there is indeed no artificial effect, which mostly reinstates confidence in the robustness of the original results and confines the observed effects to failure of being driven from any unobserved confoundings or underlying trends uninvolved with lockdown measures.

Table 6 Placebo Test Results

Variable	Estimate	Std. Error	t-value	p-value
Intercept	29.05	11.03	2.633	0.008 **
Placebo Treatment	0.76	0.48	1.592	0.111
Post	2.19	0.59	3.726	< 0.001 ***
Placebo Interaction	-0.25	0.82	-0.305	0.760
Temperature	0.66	0.025	26.114	< 0.001 ***
Humidity	-0.16	0.010	-15.874	< 0.001 ***
Wind Speed	-1.28	0.18	-7.022	< 0.001 ***
Wind Gust	-0.42	0.09	-4.824	< 0.001 ***
Pressure	0.011	0.011	1.011	0.312
Residual Std. Error		15.17 (df = 6103)		
R^2 (Adj.)		0.1582 (0.1571)		
F-statistic		143.3 (p-value < 0.001)		

Note: Robust standard errors are reported. Significance codes: *** p<0.001, ** p<0.01, * p<0.05. Placebo interaction term is not statistically significant, confirming the validity of the model.

6.5 Robustness Checks and Heteroscedasticity Robust Inference

Breusch-Pagan testing indicated the presence of very strong significance for heteroscedasticity in both the pared-back DiD model (BP = 926.12, p-value < 2.2e-16) and the complete fixed-effects DiD model (BP = 2192.2, p-value < 2.2e-16). As such, heteroscedasticity-robust standard errors (HC1 type)

enabled the derivation of acceptable, reliable statistical inferences of all estimated coefficients. The results of robust inference provided suggestive evidence of great statistical significance in the DiD interaction terms, retaining negative and statistically significant values (throughout city-day matched DiD with coefficient -15.05, robust p-value < 0.001). The robust inference strategy ensures that reported causal effects remain valid, safeguarding against potential violations of homoscedasticity.

7 Results and Analysis

This section presents the findings from our DiD analysis, divided into the following subsections: country-level aggregation analysis, city-day matching analysis (PSM DiD), and robust city-to-city matching. Each subsection has a baseline and fixed effects model with robust standard errors for heteroscedasticity.

7.1 Aggregated Country-Level DiD Analysis (Preliminary Insight)

Table 7 Aggregated Country-Level DiD Model (Without Fixed Effects)

Variable	Estimate	Std. Error	t-value	p-value
Intercept	-718.42	99.43	-7.225	< 0.001 ***
Treatment	8.43	1.57	5.365	< 0.001 ***
Post	1.82	1.59	1.144	0.253
Interaction (Treatment × Post)	-6.41	2.18	-2.934	0.003 **
Humidity	0.17	0.048	3.486	< 0.001 ***
Wind Gust	-1.33	0.34	-3.940	< 0.001 ***
Wind Speed	-1.96	0.45	-4.372	< 0.001 ***
Temperature	-0.26	0.08	-3.401	0.001 ***
Pressure	0.76	0.10	7.855	< 0.001 ***
Residual Std. Error	15.45 (df = 1030)			
R^2 (Adj.)	0.2907 (0.2852)			
F-statistic	52.77 (p-value < 0.001)			

Note: Robust standard errors are reported. Significance codes: *** p<0.001, ** p<0.01, * p<0.05.

For the first explorative step, aggregate country-level data were analyzed using a simple DiD model. The analysis consisted of meteorological control variables: temperature, humidity, wind gust, wind speed, and atmospheric pressure. According to the DiD interaction term (treatment × post), tmelt states statistically significantly lower PM2.5 levels correlating to about 6.41 $\mu g/m^3$ (p-value = 0.0034) because of the lockdown policies starting on January 31, 2020. Statistically significant were some of the meteorological variables, including Humidity (0.167, p-value < 0.001), wind gust (-1.33, p-value<0.001), wind speed (-1.96, p-value < 0.001), temperature (-0.26, p-value<0.001), and pressure (0.76, p-value<0.001). This initial finding provided weighty evidence for significant improvements in air quality attributable to strict lockdown policies, paving the way for further city-level matching analyses.

Table 8 Aggregated Country-Level DiD Model (With Fixed Effects)

Variable	Estimate	Std. Error	t-value	p-value
Intercept	-610.37	98.59	-6.191	< 0.001 ***
Treatment	11.19	1.51	7.417	< 0.001 ***
Post	-1.01	1.71	-0.591	0.554
Interaction (Treatment × Post)	-7.24	2.03	-3.563	< 0.001 ***
Pressure	0.64	0.10	6.756	< 0.001 ***
Humidity	0.24	0.05	4.855	< 0.001 ***
Wind Gust	-1.50	0.32	-4.774	< 0.001 ***
Wind Speed	-1.54	0.42	-3.692	< 0.001 ***
Temperature	1.42	0.18	8.094	< 0.001 ***
Month (Feb)	-3.88	2.96	-1.311	0.190
Month (Mar)	-11.43	2.94	-3.885	< 0.001 ***
Month (Apr)	-18.98	2.70	-7.019	< 0.001 ***
Month (May)	-31.13	3.05	-10.191	< 0.001 ***
Month (Jun)	-36.28	3.68	-9.863	< 0.001 ***
Month (Jul)	-42.52	3.94	-10.798	< 0.001 ***
Month (Aug)	-38.91	4.17	-9.339	< 0.001 ***
Month (Sep)	-44.94	3.76	-11.951	< 0.001 ***
Month (Oct)	-34.36	3.21	-10.718	< 0.001 ***
Month (Nov)	-27.67	2.80	-9.867	< 0.001 ***
Month (Dec)	-12.04	2.58	-4.672	< 0.001 ***
Residual Std. Error	14.12 (df = 1019)			
R^2 (Adj.)	0.4136 (0.4027)			
F-statistic	37.83 (p-value < 0.001)			

Note: Fixed effects included for month. Robust standard errors are reported. Significance codes: *** p<0.001, ** p<0.01, * p<0.05.

A fixed-effects version of the aggregated country-level DiD model, with month-specific fixed effects, was estimated to control more explicitly for time-invariant heterogeneity and time-varying seasonal effects. In this fixed-effects specification, the interaction term remained negative, statistically significant (-7.24, p-value<0.001), and consistent with the baseline aggregated DiD model. Moreover, the meteorological variables: Humidity (0.24, p-value<0.001), wind gust (-1.50, p-value<0.001), wind speed (-1.54, p-value<0.001), temperature (1.42, p-value<0.001), and pressure (0.64, p-value<0.001) maintained significance. The monthly fixed effects indicated large and significant seasonal variations, which supported the justification for the inclusion of month fixed effects

7.2 City-Day Matched PSM+DiD Model Results

Using matching city-day DiD methods which employed propensity score matching based on log population and log GDP per capita our city-day refined estimates were highly informative for determining direct causal estimates at the city-day level.The interaction term for the DiD is consistently strong and

Table 9 City-Day Level Matching Difference in Differences Results (Without Fixed Effects)

Variable	Estimate	Std. Error	t-value	p-value
Intercept	−56.74	14.39	-3.94	$8.07e-05$ ***
Treatment	17.83	0.50	35.99	$< 2e-16$ ***
Post	3.07	0.58	5.24	$1.62e-07$ ***
Temperature	-0.07	0.04	-1.82	0.0692 .
Humidity	-0.007	0.012	-0.58	0.5614
Wind Speed	-0.27	0.06	-4.47	$8.04e-06$ ***
Wind Gust	-1.10	0.06	-18.58	$< 2e-16$ ***
Pressure	0.10	0.01	7.11	$1.22e-12$ ***
Dew Point	-0.006	0.05	-0.12	0.9069
Treatment × Post	-14.19	0.82	-17.34	$< 2e-16$ ***
Observations		12,224		
Adjusted R^2		0.151		

Note: Robust standard errors (HC1 type) are reported. Significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Fixed effects are not included in this specification.

was thus significantly different to the mean, indicating some level of support for significantly decreased PM2.5 levels (regardless of how that interacts in space) (coefficient = -14.19,p-value <0.001).The control variables of temperature (-0.073,p-value = 0.0058), wind speed (-0.27,p-value<0.001), wind gust (-1.10,p-value<0.001), atmospheric pressure (0.10,p-value<0.001) are found to be significant. The matching city-day model was well explained by several factors(Adjusted R^2 = 0.1571), thereby affording further support to the cause-effect chain of the lockdown on a city-day resolution.

7.3 Matched City-Day DiD Model with Fixed Effects

Table 10 City-Day Level Matching Difference in Differences Results(With Fixed Effects)

Variable	Estimate	Std. Error	t-value	p-value
Intercept	−236.85	22.69	-10.44	< 0.001 ***
Post	4.24	0.60	7.07	< 0.001 ***
Temperature	1.37	0.05	26.94	< 0.001 ***
Humidity	0.09	0.01	7.79	< 0.001 ***
Wind Speed	-0.15	0.06	-2.78	0.005 **
Wind Gust	-1.00	0.06	-17.21	< 0.001 ***
Pressure	0.29	0.02	13.18	< 0.001 ***
Treatment × Post	-15.05	0.75	-20.01	< 0.001 ***
City Fixed Effects	Yes			
Month Fixed Effects	Yes			
Observations		12,224		
Adjusted R^2		0.323		

Note: Robust standard errors (HC1 type) are reported. Significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. City and month fixed effects are included in the model but omitted here for brevity.

Additionally, the matching city-day DiD model considered city and month fixed effects, expressing the need to explicitly control for city-specific time-variant heterogeneity as regards PM concentrations. Once fixed effects were admitted, the DiD interaction term still was strongly significant and pointed to a substantially decreased PM2.5 concentrations (-15.05 $\mu g/m^3$, with robust p-value<0.001). Similar relationships were also preserved for meteorological controls (temperature 1.37, humidity 0.09, wind speed -0.15, wind gust -1.00, and pressure 0.29, with all p-values<0.05). The matching city-day model has large explanatory power (Adjusted R^2 = 0.3017). Altogether these findings illustrated the robustness of the treatment estimates when controlling for city and month fixed effects and importantly underlined the credibility of the DiD results post-coefficient adjustment for any unobserved inconsistencies.

7.4 One-to-One City Matched DiD Analysis (Robustness Check)

Table 11 One to One City Matched Difference in Differences Results (Without Fixed Effects)

Variable	Estimate	Std. Error	t-value	p-value
Intercept	−278.72	23.20	-12.01	$< 2e-16$ ***
Treatment	17.37	0.49	35.12	$< 2e-16$ ***
Post	5.26	0.60	8.82	$< 2e-16$ ***
Temperature	-0.06	0.03	-2.13	0.0336 *
Humidity	-0.042	0.011	-3.72	0.0002 ***
Wind Speed	-0.28	0.06	-4.57	$4.94e-06$ ***
Wind Gust	-1.01	0.06	-16.25	$< 2e-16$ ***
Pressure	0.32	0.02	14.17	$< 2e-16$ ***
Treatment × Post	-16.47	0.83	-19.93	$< 2e-16$ ***
Observations		12,148		
Adjusted R^2		0.151		

Note: Robust standard errors (HC1 type) are reported. Significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Fixed effects are not included in this specification.

In order to confirm robustness further, a simpler matched analysis was done, in one-to-one city matching, where cities were matched once according to their population and GDP per capita criteria—a fixed throughout the entire analysis period. The DiD interaction was still significant at a large negative estimate (-16.47, with p < 0.001 threshold). Meteorological variables such as temperature (-0.06), humidity (-0.042), wind speed (-0.28), wind gust (-1.01), and pressure (0.32) displayed expected signs and significance levels to demonstrate the models' stability even with this fixed city-pair approach, with a fairly good explained variance (Adjusted R^2 = 0.151).

7.5 Matched One-to-One DiD with Fixed Effects

Table 12 One to One City Matched Difference in Differences Results (With Fixed Effects)

Variable	Estimate	Std. Error	t-value	p-value
Intercept	−224.17	23.29	-9.62	$< 2e-16$ ***
Post	4.27	0.61	7.04	$2.05e-12$ ***
Temperature	1.04	0.05	22.20	$< 2e-16$ ***
Humidity	0.053	0.012	4.25	$2.12e-05$ ***
Wind Speed	-0.18	0.06	-3.13	0.00175 **
Wind Gust	-1.08	0.06	-17.86	$< 2e-16$ ***
Pressure	0.28	0.02	12.49	$< 2e-16$ ***
Treatment × Post	-15.37	0.76	-20.15	$< 2e-16$ ***
City Fixed Effects		Yes		
Month Fixed Effects		Yes		
Observations		12,115		
Adjusted R^2		0.304		

Note: Robust standard errors (HC1 type) are reported. Significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. City and month fixed effects are included in the model but omitted here for brevity.

The model was then succeeded by an additional matched city-level fixed-effects DiD model that further improved estimates by directly controlling for time-invariant city-specific characteristics and seasonal effects. In this stricter specification, the DiD interaction term was still significant (-15.37, with p < 0.001 threshold), providing robust evidence regarding reductions in PM2.5 concentrations induced by the lockdowns. The meteorological variables remained significant, with temperature (1.04), humidity (0.053), wind speed (-0.18), wind gust (-1.08), and atmospheric pressure (0.28), significantly influencing PM2.5 (p < 0.05). Since the fixed effects for several cities were frequently significant, considerable baseline differences across locations were apparent. The model's explanatory power was greatest (Adjusted R^2 = 0.304), supporting the robustness and credibility of this refined specification within the confines of causal estimates.

7.6 Heterogeneous Effects Analysis

In addition to its primary results, this study also assesses heterogeneous treatment effects in order to understand how the effect of lockdown measures might differ between cities and at different times. An inquiry into these heterogeneous effects allows tailoring of policy recommendations and an understanding of conditions under which lockdown policies could yield larger environmental benefits.

7.6.1 City-Level Heterogeneity

Panel fixed effects at city level used in matched treated-to-control pair setup and other city matched Design models indicated possible baseline differences in PM2.5 levels across cities. In Italy, cities such as Milan, Naples, Brescia, Turin, and Parma were persistently revealed to have higher PM2.5 levels at the baseline through the signs of significantly positive city fixed effects (Milan: 13.96 µg/m³, Naples: 11.50 µg/m³, Turin: 10.95 µg/m³; by contrast, certain Japanese cities, including Shizuoka, Chiba, Kanazawa, Toyama, and Tokyo showed significantly lower PM2.5 concentrations at the baseline, Shizuoka: -18.17 µg/m³, Kanazawa: -16.95 µg/m³, Toyama: -17.43 µg/m³). Such city-specific fixed effects explain a massive heterogeneity across urban pollution baselines, most likely brought about by diverse industrial profiles, economic forms, and urban characteristics.

Table 13 City-Level and Temporal Heterogeneity in $PM_{2.5}$ Levels

Variable	Estimate	Std. Error	t-value	p-value
City Fixed Effects				
Brescia	8.96	1.20	7.50	$< 2e-16$ ***
Chiba	-15.20	1.22	-12.50	$< 2e-16$ ***
Florence	-2.99	1.18	-2.53	0.011 *
Hiroshima	-10.01	1.31	-7.64	$2.42e-14$ ***
Kanazawa	-16.95	1.22	-13.92	$< 2e-16$ ***
Livorno	-6.95	1.19	-5.86	$4.87e-09$ ***
Milan	13.96	1.18	11.79	$< 2e-16$ ***
Modena	10.34	1.18	8.73	$< 2e-16$ ***
Nagano	-12.11	1.25	-9.72	$< 2e-16$ ***
Nagoya	19.50	2.68	7.28	$3.51e-13$ ***
Naples	11.50	1.19	9.68	$< 2e-16$ ***
Osaka	-10.80	1.23	-8.78	$< 2e-16$ ***
Parma	9.69	1.19	8.14	$4.19e-16$ ***
Shizuoka	-18.17	1.35	-13.48	$< 2e-16$ ***
Tokyo	-11.66	1.23	-9.52	$< 2e-16$ ***
Toyama	-17.43	1.21	-14.37	$< 2e-16$ ***
Turin	10.95	1.18	9.25	$< 2e-16$ ***
Month Fixed Effects				
February	-14.47	1.06	-13.62	$< 2e-16$ ***
March	-16.20	1.11	-14.62	$< 2e-16$ ***
April	-24.91	0.96	-25.91	$< 2e-16$ ***
May	-36.55	1.05	-34.84	$< 2e-16$ ***
June	-39.50	1.20	-33.06	$< 2e-16$ ***
July	-44.99	1.27	-35.51	$< 2e-16$ ***
August	-41.05	1.32	-31.02	$< 2e-16$ ***
September	-45.70	1.25	-36.47	$< 2e-16$ ***
October	-35.14	1.13	-31.10	$< 2e-16$ ***
November	-31.38	1.03	-30.41	$< 2e-16$ ***
December	-13.73	0.99	-13.81	$< 2e-16$ ***

Note: Robust standard errors (HC1 type) are reported. Significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Estimates represent city and seasonal heterogeneity in $PM_{2.5}$ levels.

7.6.2 Temporal (Seasonal) Heterogeneity

Meanwhile, specific month effects indicated significant seasonal fluctuation on PM2.5 concentrations in all matched city-level models. For example, models pointed out consistently lower levels of PM2.5 in warm months (May through September) in comparison to cooler months. The most substantial winter times reductions occurred from -36.55 $\mu g/m^3$ in May to -45.70 $\mu g/m^3$ in September, plausibly aligning with seasonal variations related to meteorological conditions. This suggests that the effect of lockdown measures on air quality may have been shaped by seasonal influences and the effectiveness of the measures varied with time, based on meteorological conditions, such as temperature, and wind flow.Considerable heterogeneity at city and seasonal levels reveals lockdown measures have not operated uniformly on air quality. With their higher levels of background air pollution and relatively elevated industrial activity, cities like Milan and Turin might have seen a greater reduction due to the halting or slowing down of economic and industrial interactions. Again seasonal differences may allow the timing of the lockdown to be instrumental in determining its effectiveness toward improving air quality.

8 Discussion and Policy Implications

8.1 Key Insights and Contributions

This piece of research offers substantial causal evidence of the profound effect of lockdown measures on urban air quality, explicitly indicating the reductions in concentrations of PM2.5. The analysis combines Propensity Score Matching (PSM) with Difference-in-Differences (DiD) analysis at three levels of aggregation: country level, city-day matched, and one-to-one city matched. The analysis constantly identifies massive reductions in PM2.5 levels associated with strict lockdown policies enforced in Italy compared with moderate policy restrictions in Japan. Crucially, the detrimental DiD consequences statistically persisted for all methodological specifications throughout this study, defining a stark causal relationship between the restrictions due to lockdown and an amelioration of air quality. The level of heterogeneity was also driven by city economic structures and demographic characteristics, where cities such as Italy's industrialized and populous cities offered greater reductions in absolute terms concerning milieu. However, the strength of the lockdown effects was also influenced by clear seasonal patterns, indicating that meteorological conditions are important modulators of emission reduction measures' effectiveness. These insights, therefore, strengthen the hypothesis that human economic activities constitute a major causal factor into the dynamic of urban air pollution impacts and have substantial environmental paybacks by aiming to decrease emissions from industrial and transport sources considerably. This study is an important addition to the growing literature on the environmental consequences of COVID-19 lockdowns, using a rigorous methodological framework involving city-level matching and Difference-in-Differences (DiD) techniques. Most previous studies relied on aggregated country-level data or descriptive analyses, while this paper has based itself on a city-day matched dataset that establishes causal inference frameworks by tackling biases from unobserved confounders and pre-existing trends.Building on Ma et al. (2024) and Levelt et al. (2022), which discussed the importance of meteorological normalization and satellite-space data, the study adds time-varying meteorological controls to improve its estimates of lockdown impacts on PM2.5 levels.With granular city-level data and matching techniques, causal inferences are much more precise and further advance methodological discourse.Consistent with Nguyen et al. (2022) and Addas and Maghrabi (2022), who reported regional variability in altering air quality, the study formalizes heterogeneous impacts by urbanization and over time.Highly urbanized Italian cities showed larger reductions in PM2.5, which reflect effects of baseline pollution levels and more stringent restrictions on mobility. This feature reinforces broad findings from Liu et al. (2021) and Dahu et al. (2022) and adds credence to the importance of local-level analysis in environmental studies. Furthermore, this study addresses the methodological diversity issues outlined in the literature, by deploying, among others, One-to-One City Matching and Aggregated Country-Level Data for robustness checks.Alternatively giving credence to the concurrent expectations of well-founded and central declarations, the evidence in support of a common environmental equilibrium provides the grounds for the design of evidence-informed policies with competing claims of environmental sustainability on the one hand and public-health appeal on the other, giving actionable advice to city planning and regional analysis.

8.2 Policy Implications

These results carry important implications for the formulation of environmental and urban policies. Firstly, the analysis demonstrates that stringent, targeted emissions control strategies, like the activity reductions enforced during lockdowns due to the COVID-19 pandemic, can rapidly and significantly improve urban air quality. Hence, policymakers should give priority to targeted and carefully planned interventions to reduce industrial emissions and the activities of vehicles, especially during periods of pollution spikes. The heterogeneity revealed within the cities also includes the implication that environmental and industrial policies need to be tailored to the discrete nature of individual regions. More industrialized or populous cities commonly retain a higher baseline pollution level, hence making room for aggressive and targeted emission reduction policies. Integration of such insight into urban planning frameworks will help optimize effectiveness by considering local economies and demography. Finally, with air quality being inherently tied to public health, an integrated public health and environmental approach should be embraced by all policymakers. Armed with lockdown-related improvement benchmarks, authorities can ensconce sustainable urban designs that balance an ever-burdening growth economy and public health and environmental goals.

8.3 Limitations and Future Research

This is a significant paper because it tries to integrate the positive elements with a number of limitations. Locking the environment up is one-time shock, thus its impact would last only a couple of hours. This cannot be considered a long-term environmental policy since its results would only be short-lived; one can expect no improvement of the air quality for a long time. Research should also consider possible rebound effects or other lasting changes once the lockdowns lift. Besides, the locking-up approach already takes into account demographic, economic, and meteorological variables while pairing up. Thus, there might be other unmeasured variables that might induce some level of residual confounding that could relate to the industrial definitions or the way people commute, thus affecting trends on air pollution. The other point worth mentioning is that the analysis made in this study was conducted in two very developed urban centers; hence, generalization would become too complicated for rural regions or cities facing completely different socioeconomic conditions. Therefore, future works must take these constraints into account by broadening the temporal horizon in their assessment in relation to the longer-lasting effects and responses, expand the spatial extent for comparative studies, and develop very detailed data about particular sectoral or mobility aspects that could strengthen causation and relevance for policy-making.

8.4 Broader Implications

This study contributes other important general findings related to environmental policies and urban sustainability apart from the obvious air quality aspect. There is, to start with, the immediate enhancement of air quality in the city, which points toward a potential for future benefits for environment through the deliberate reduction of emissions. Insights drawn from this experience could teach valuable lessons toward achieving mitigation strategies for climate change, in particular, pointing to large reductions of emissions that can yield rapid environmental change. In this sense, this study highlights that more attention should go into the integration of the principles of sustainable urban development in environmental management processes. Thus, insights that come from this study could guide urban planners and policymakers toward creating sustainable and resilient urban environments in cities by prioritizing cleaner industrial processes and efficient public transportation systems. In addition, increased public awareness generated through visible improvement of air quality in lockdowns is also a great opportunity for sustained behavioral change and more support for stronger environmental policies. Policymakers should grab the increased awareness that could bring long-term engagement of urban populations with environmental matters.

9 Conclusion

This study systematically quantified the causal effect of stringent lockdown policies to contain COVID-19 on PM2.5 air pollution with the use of robust econometric methodologies like Difference-in-Differences combined with Propensity Score Matching, both at the city-day level and aggregated levels. Across all model specifications and robustness checks, the results remained consistent, showing significant substantial reductions in urban PM2.5 levels due to lockdown interventions, particularly in dense, economically active cities. The reliability and causal robustness of the find-

ings were confirmed by the rigor of diagnostics such as multicollinearity tests, covariate balance tests, parallel trend tests, placebo falsification tests, and robust inference methods. From a policy standpoint, there is compelling evidence showing how quickly and effectively selected reductions in certain human activities lead to payoffs in environmental improvement and human health. These findings not only highlight the potential for carefully designed environmental policies but also showcase that equally critical are localized and economically targeted measures. Notwithstanding some limitations regarding the time frame and generalizability of our results, evidence underscores solutions to policy-induced interventions that can potentially ameliorate urban air quality, providing further avenues toward sustainable environmental management purposely embedded in urban settings. Moving forward, they can widen the tent for further studies stretching these lessons across different temporal, spatial, and economic contexts, integration laying foundation for more informed policy decision-making toward sustainable operations.

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